

Global CO₂

Emissions, Regional

Shifts, Per Capita

Inequalities

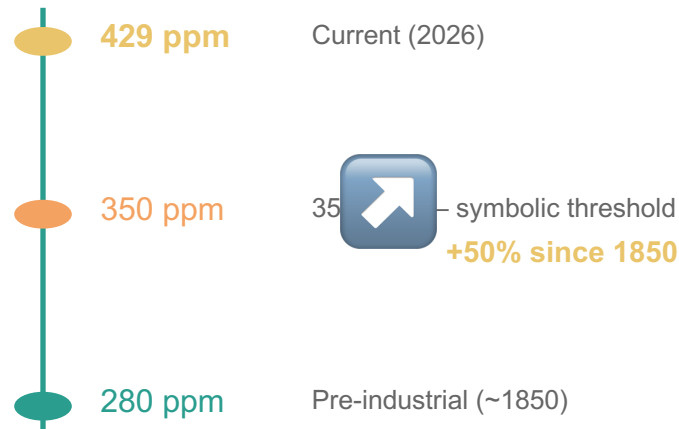
•• Data-Driven Analysis · 1750–2024

429

ppm

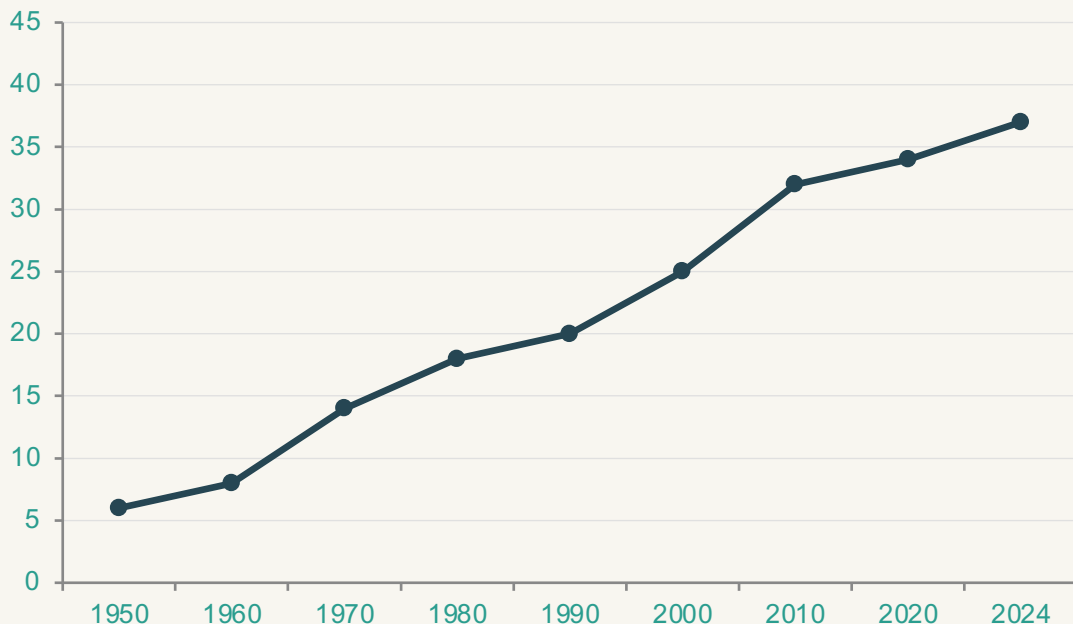
Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

Current level
429
ppm
1.5× pre-industrial level



- Pre-industrial baseline: **~280 ppm** • Current level is **1.5× greater** than natural ice-age increase (which took thousands of years)
- Rapid rise occurred in **~200 years** — linked to burning of coal, oil, and natural gas

Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes



1950

0

~6 Bt

1990

0

~20 Bt

2024

4

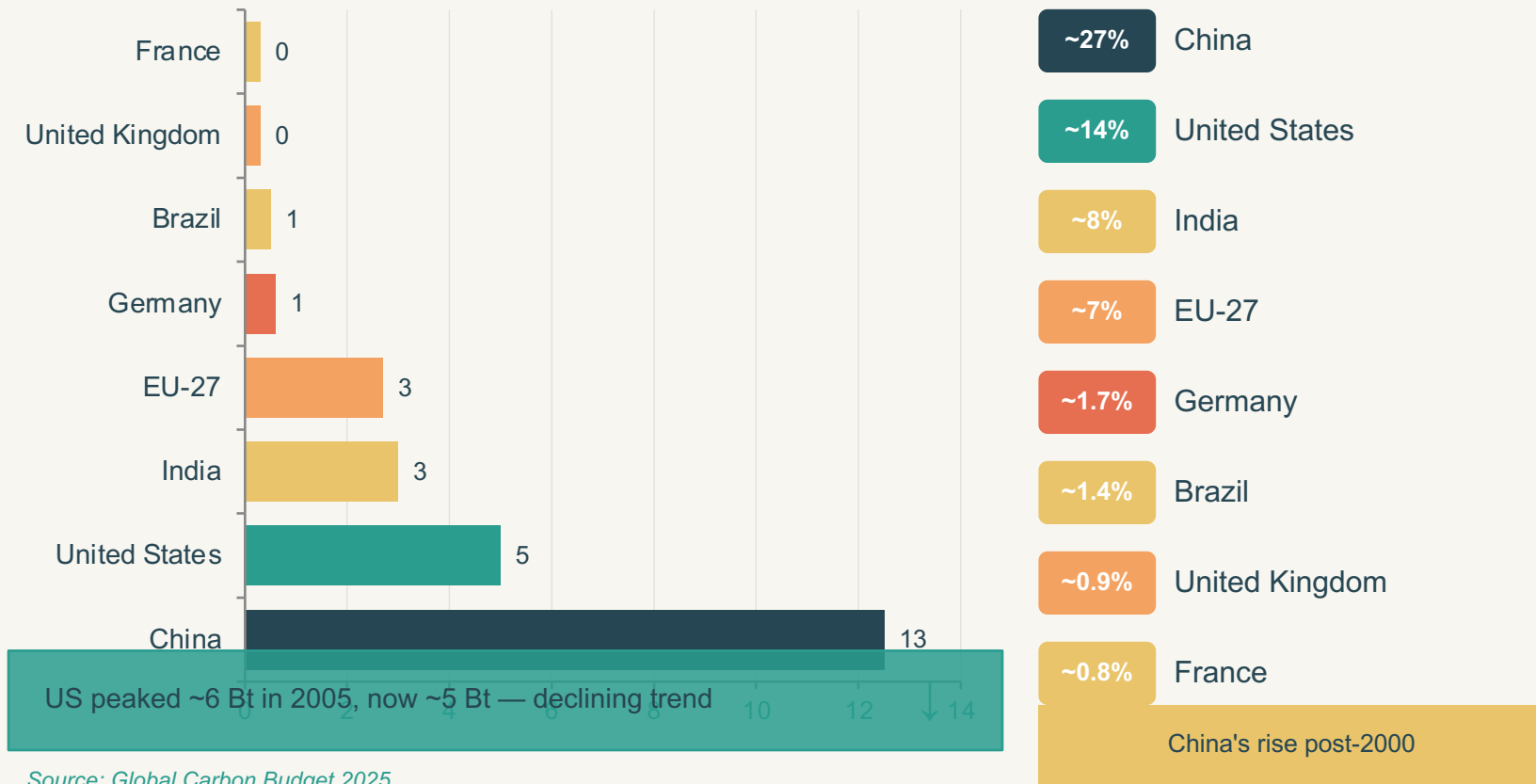
35+ Bt

Before the Industrial Revolution, CO₂ emissions were negligible. Growth remained slow until the mid-20th century.

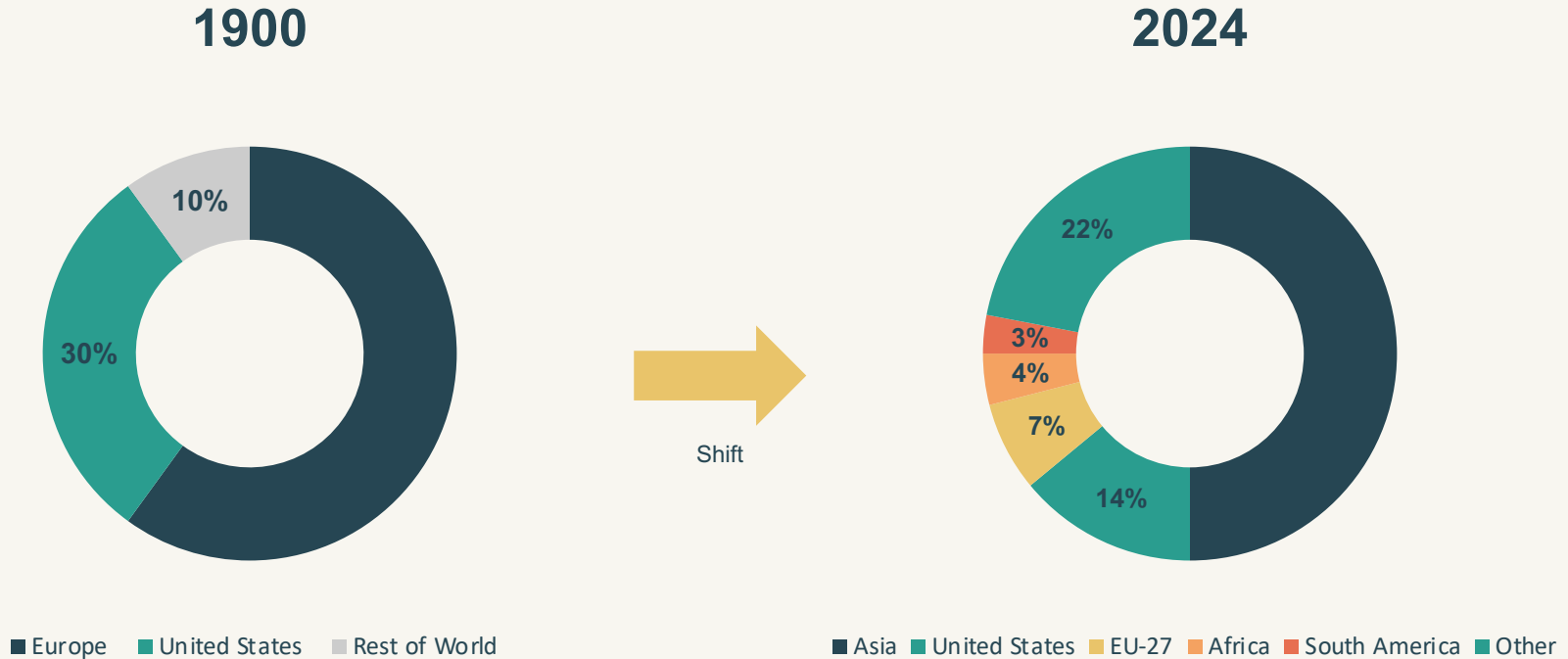
While emissions growth has slowed in recent years, they have not yet peaked.

Source: Global Carbon Budget 2025

China Now Emits Over 12 Billion Tonnes — More Than Double the United States



Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third



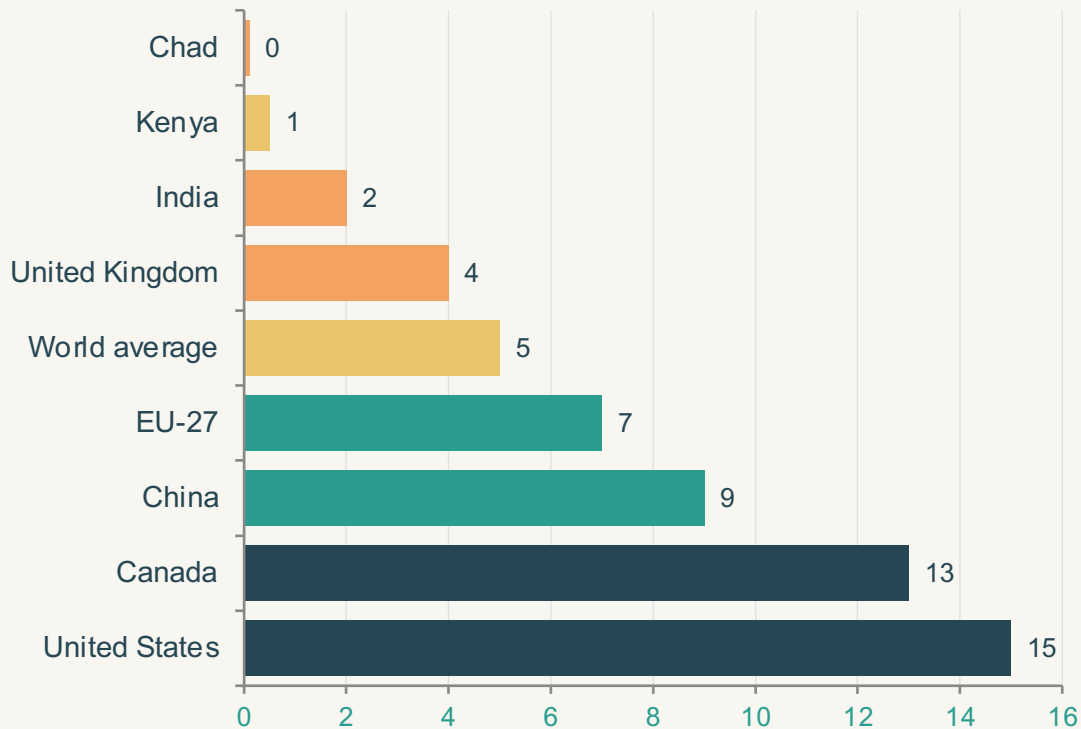
Europe + US: >90%

US + Europe: <33%

Asia (60% of world pop.) produces ~50% of emissions

Africa + South America: ~7% (vs. aviation & shipping)

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity



150× Disparity

The average American emits in under two days what the average person in Chad emits in an entire year.

See figure_1.jpg for per capita line chart over time

High-income nations: ~10× the global average per capita

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

United States

Declining (-1% to -5%)

Much of Europe

Declining (-1% to -5%)

Russia

Increasing (+2% to +10%)

SE Asia

Increasing (+2% to +10%)

Parts of Africa

Increasing (+2% to +10%)

South America: Mixed patterns

Color Legend



Blue: Negative change (-1% to -5%)



Orange: Positive change (+2% to +10%)



Red: Large increase (>10%)

Map Overview (figure_4.jpg)

Shows 2024 annual % change in CO₂ emissions globally

- Blue regions: US & Europe
- Orange/red: Russia, SE Asia, Africa
- South America: mixed results

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Lower Per Capita Emissions

France

~70% nuclear
High renewables

~4 t CO₂/person

United Kingdom

High renewables
Declining coal

~4 t CO₂/person

Higher Per Capita Emissions

Germany

High fossil (coal gas), nuclear phase-out

~8 t CO₂/person

Netherlands

High fossil fuel dependence

~8–9 t CO₂/person

Key Insight: Countries with similar living standards can have very different per capita emissions depending on their energy sources.

Low Emission Sources

Nuclear & Renewables dominate

High Emission Sources

Coal, Gas & Oil dominate

Source: IEA · Global Carbon Budget 2025

Per capita CO₂ = total emissions / population

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

- 1 Global emissions exceed 35 Bt and have not yet peaked
- 2 China alone emits >25% of global total (~12.5 Bt), more than double the United States (~5 Bt); India (~3 Bt) is the third-largest and rising fast
- 3 The geographic center of emissions has shifted dramatically: from >90% in Europe and the US in 1900 to Asia producing ~50% today, while the US and Europe combined are below one-third
- 4 Per capita inequalities remain extreme — a 150x gap separates the highest emitters (US, Australia at ~15 t/person) from the lowest (Chad, Niger at ~0.1 t/person)
- 5 Policy and energy mix choices — not just prosperity — determine per capita outcomes (e.g., France vs. Germany)